Shubham Pande

Personal webpage | LinkedIn | Google Scholar   Email: shubhampande142@gmail.com | Mobile: (+91)8698491057

Research Interest

Semiconductor Device Physics, Compact Modeling of Emerging Devices, Novel Memory Concepts, Neuromorphic Computing

Education

Indian Institute of Technology, Madras 2019 – 2025
Direct PhD (MS + PhD) ***CGPA - 9.41***

SGGSIE&T, Vishnupuri, Nanded, Maharashtra 2012 – 2016
B.Tech in Electronics and telecommunication ***CGPA - 8.02***

Professional Experience

Intel Technology, India Pvt. Ltd. June 2024 – Present
Device Modeling Engineer *Hyderabad, India*

- Contributing to the development of Process Design Kits for Intel's upcoming 14A/10A technology node
- Working on device and interconnect modeling to support advanced process development
- Collaborating closely with internal teams—including Technology Development, Design Rule, and Circuit Design groups—to gather inputs and define design targets

Research Experience

ReRAM Memory Technology for Neuromorphic Computing – Design, Modeling, and Applications (PhD) 2020 – 2025
Advisors: Prof. Anjan Chakravorty & Prof. Bhaswar Chakrabarti *IIT Madras, India*

- Performed device-level simulation, compact modeling, and electrical characterization of ReRAM devices
- Investigated electrothermal effects in ReRAM crossbar and analyzed their impact on analog in-memory computing
- Proposed novel synaptic devices and circuit architectures for hardware implementation of bio-inspired learning rules

NeoHebbian Synapses for Online Learning in Spiking Neural Networks (Fulbright Fellow) Aug 2022 – July 2023
Advisor: Prof. Dmitri Strukov *UC Santa Barbara, USA*

- Worked on Online Learning of Spiking Neural Networks
- Developed novel synaptic designs for the hardware implementation of the eligibility-propagation algorithm

Modeling Electrothermal Effects in SiGe HBTs 2019 – 2020
Advisor: Prof. Anjan Chakravorty *IIT Madras, India*

- Investigated electrothermal effects in semiconductor devices to improve reliability and prevent failures
- Developed techniques for co-optimizing thermal and electrical performance
- Built physics-based compact models for static and dynamic temperature behavior in HBTs and LEDs

Development of Low-Cost BiCMOS Process 2018 – 2019
Advisors: Prof. Anjan Chakravorty & Prof. Nihar Mohapatra (IITGN, India) *IIT Madras, India*

- Improved RF and DC performance of parasitic BJTs in SCL's 180nm CMOS technology by implementing minimal-mask process modifications, enabling a cost-effective BiCMOS integration
- Executed comprehensive TCAD process and device simulations to optimize transistor performance and validate design enhancements
- Gained expertise in TCAD calibration, electrothermal simulations, electrical characterization, parameter extraction, and DC/RF layout design

Publications

Related to ReRAM for neuromorphic computing

Journal publications

* **Shubham Pande**, T. Bhattacharya, S.S. Bezugam, E. Wlazlak, and D. Strukov, "Accelerating Neuromorphic Hardware Training with ReRAM NeoHebbian Synapses", Under review
* **Shubham Pande**, S. Balanethiram, B. Chakrabarti, and A. Chakravorty, "A Physics-Based Compact Model for Thermal Resistance in Resistive Random-Access Memories", Solid-State Electronics, vol. 204, 2023
* Masud R. Sk, **Shubham Pande**, F. Müller, Y. Raffel, M. Lederer, L. Pirro, S. Beyer, K. Seidel, T. Kämpfe, S. De, and B. Chakrabarti, "Fixed Charges at the HfO_2/SiO_2 Interface: Impact on The Memory Window of FeFET", Memories - Materials, Devices, Circuits and Systems, Vol. 4, 2023
* L.H. Reddy , **Shubham Pande**, T. Roy, E.M. Vogel, A. Chakravorty, and B. Chakrabarti, "A SPICE compact model for forming-free, low-power graphene-insulator-graphene ReRAM technology", Emergent Materials, 4, 1055-1065, 2021

Conference publications

* T. Bhattacharya*, S.S. Bezugam*, **Shubham Pande***, E. Wlazlak, and D. Strukov, "ReRAM-Based NeoHebbian Synapses for Faster Training-Time-to-Accuracy Neuromorphic Hardware", 2023 International Electron Devices Meeting (IEDM), 2023, pp. 1-4, *Equal contribution
* **Shubham Pande**, M.R. Sk, Y. Raffel, M. Lederer, K. Seidel, A. Chakravorty, S. De, and B. Chakrabarti, "FeFET based LIF Neuron with Learnable Threshold and Time Constant", IEEE Device Research Conference (DRC), 2024
* S. Roy, **Shubham Pande**, M.R. Sk, E. Bhattacharya, and B. Chakrabarti, "Demonstration of an Oscillatory Neuron using SiO_x-based Memristive Switches", IEEE Device Research Conference (DRC), 2024
* **Shubham Pande**, B. Chakrabarti, and A. Chakravorty, "On the Prevalence of Row Hammer Attacks in FeFETs based Memory Systems", IEEE Electron Device Technology Meeting (EDTM), 2024
* V.V. Gabriel, **Shubham Pande**, B. Chakrabarti, and A. Chakravorty, "A Low Power LIF Neuron Circuit with Programmable Time Constant", accepted at 18^{th} ACM International Symposium on Nanoscale Architecture (Nanoarch), 2024
* **Shubham Pande**, B. Chakrabarti, and A. Chakravorty, "Electro-thermal Confinement Leads to Low Power ReRAM Operation", XXII International Workshop on Physics of Semiconductor Devices - 2023
* **Shubham Pande**, B. Chakrabarti and A. Chakravorty, "Leveraging MOSFET V_{th} Temperature Sensitivity for Memory Application", Non-volatile Memory Technology Symposium (NVMTS), imec, Belgium, 2023
* **Shubham Pande**, S. Balanethiram, B. Chakrabarti, and A. Chakravorty, "2D-Motion Detection using SNNs with Graphene-Insulator-Graphene Memristive Synapses", 29^{th} IEEE International Conference on Electronics, Circuits, and Systems (ICECS), 2022
* S. Roy, **Shubham Pande**, B. Chakrabarti, and E. Bhattacharya, "A SiOx Resistive Memory with Low Operating Voltages, Gradual Set/Reset Operation, and High On-State Non-linearity", MRS spring 2022

Related to electro-thermal modeling

Journal publication

* Nidhin K., **Shubham Pande**, S. Yadav, S. Balanethiram, D. R. Nair, S. Frégonèse, T. Zimmer, and A. Chakravorty, "An Efficient Thermal Model for Multifinger SiGe HBTs Under Real Operating Condition" IEEE Transactions on Electron Devices 67, 2020

Conference publications

* **Shubham Pande**, Nidhin K., S. Yadav, S. Balanethiram, and A. Chakravorty, "Thermal Impedance Model for Multifinger SiGe HBTs", IEEE Electron Device Technology Meeting (EDTM), 2024
* S. Barman, **Shubham Pande**, S. Balanethiram, T. Zimmer, S. Fregonese and A. Chakravorty, "Systematic Extraction of Thermal Network Parameters in Multi-Finger SiGe HBTs," IEEE BiCMOS and Compound Semiconductor Integrated Circuits and Technology Symposium (BCICTS), 2024.
* **Shubham Pande**, Nidhin K., K.P. Navaneeth, S. Balanethiram, and A. Chakravorty, "Analytical Modeling of Junction Temperature and Thermal Resistance in LED Structure", XX^{th} International Workshop on Physics of Semiconductor Devices - 2019

Related to development of low-cost BiCMOS process

Conference publications

* **Shubham Pande**, S. Balanethiram, A. K. Singh, M. Gupta, B. Umapathi, H. S. Jatana, N. R. Mohapatra, and A. Chakravorty, "Development of Low-Cost Silicon BJT Technology and Modeling", 5^{th} IEEE International Conference on Emerging Electronics (ICEE), 2020
* S. Balanethiram, **Shubham Pande**, A. K. Singh, B. Umapathi, H. S. Jatana, N. R. Mohapatra, and A. Chakravorty, "Development of Low-Cost Silicon BiCMOS Technology for RF Applications", IEEE Conference on Modeling of Systems Circuits and Devices (MOS-AK India), 2019

Patents

- **Shubham Pande**, B. Chakrabarti, and A. Chakravorty, "A ReRAM device for low power operation and a method for fabrication thereof", Indian patent granted, patent no: 566655, application no: 202341081200
- **Shubham Pande**, B. Chakrabarti, and A. Chakravorty, "Apparatus for Storing Information on a Semiconductor Device, and Method Thereof", Indian patent filed, application no. 202341055188, 17 Aug 2023

Book Chapters

- B. Chakrabarti, **Shubham Pande**, S. Roy, and M. Sk. Rana, "Oxygen vacancy based resistive memory for storage and computing applications", in press
- Sourav De, **Shubham Pande**, Masud Rana Sk, Ankush Chattopadhyay, Yannick Raffel, B. Chakrabarti, and Tuo-hung Hou , "Hafnium Oxide-Based Ferroelectric Memories: Applications and Future Prospects", in press

Technical Skills

Device simulator : Synopsys Sentaurus TCAD (Sprocess, SDE, Sdevice), COMSOL Multiphysics
SPICE simulators: Cadence - Spectre, QucsStudio, LTspice
Prog. languages : Verilog-A, Python (Libraries used: Matplotlib, NumPy, SciPy, Pandas, Scikit-learn), LaTeX
Software tools : MATLAB, Klayout, Origin Lab, snnTorch, JMP, NeuroSIM
Exp. tools : B1500 parametric analyzer, Manual probe station

Fellowships

Fulbright-Nehru Doctoral Research Fellowship Aug 2022 – July 2023

- Awarded by the The United States-India Educational Foundation
- Conducted research at the University of California, Santa Barbara, under the guidance of Prof. Dmitri Strukov.

Prime Minister's Research Fellowship (PMRF) 2020 - 2024

- Awarded by the Government of India to support outstanding research talent and innovation.

Central Sector Scholarship 2012-2016

- Provided financial assistance for higher education to meritorious students by the Government of India.